Application of Artificial Intelligence in German Municipal Wastewater Facilities

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Abstract: This paper investigates the current state, perceived potential, and barriers to the adoption of AI-based solutions in German municipal wastewater treatment plants, based on an online survey of 105 respondents and ten expert interviews. AI adoption remains limited despite high awareness of its potential: respondents expect the greatest benefits in operational safety, process stability, regulatory compliance, and improved documentation, and existing monitoring systems provide favorable conditions for AI integration, whereas concerns regarding technical reliability, system integration, costs, and IT security hinder wider deployment. Bivariate analyses reveal that awareness increases with plant size, whereas implementation remains consistently low across all facility categories; the perceived benefits are largely consistent regardless of plant size or prior AI experience, with AI-supported detection considered most valuable for process-critical events. To narrow this gap, the paper derives recommendations pertaining to pilot projects, knowledge transfer, and economically viable, easily integrated solutions, and outlines future research directions.

Keywords: Artificial Intelligence in Wastewater Management, AI Adoption Barriers, German Municipal Wastewater Treatment Plants

1 Introduction

Digitalization is increasingly transforming wastewater management. Rising requirements pertaining to efficiency, resource conservation, and the protection of water bodies, an increasing volatility within the sewer network, more frequent heavy rainfall events, and a growing shortage of skilled personnel render data-based approaches increasingly important. Methods of artificial intelligence (AI), in particular, offer considerable potential to optimize operational processes, for instance through improved forecasting, adaptive control, or automated anomaly detection. Particularly since the effectiveness of the European wastewater sector has been called into question [BJBB22], [Euro23].

Within environmental information systems, AI moreover opens up novel possibilities to systematically evaluate large volumes of data from measurement, control, and monitoring systems for operational decision-making. Camera-based AI solutions, for instance, may serve as an additional online monitoring method for the detection of anomalies such as discolorations, foaming, and oil films [APBS24], [ICMB23], [WCLG23].

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Notwithstanding this potential, the extent to which AI applications are actually deployed in German wastewater facilities has received scant empirical attention to date. Against this background, a nationwide online survey with more than one hundred participating representatives of municipal wastewater treatment plants was conducted, complemented by expert interviews; the present paper summarizes the principal findings and depicts the current status of AI deployment together with central potentials and barriers.

2 Related Work

A substantial body of international research on AI applications in wastewater management exists. Hocaoglu et al. [HRGM25] differentiate three principal classes of AI-based models: predictive models, detective models, and AI-based optimization models.

Predictive models are concerned with the forecasting of relevant parameters, such as wastewater quality parameters and emissions from treatment plants (e.g. [ShAH20]). Detective models address the identification of faults and the detection of anomalies, whose principal causes are illicit discharges, defects within the treatment system, and leakages in the sewer network. Camera-based AI solutions for anomaly detection have been examined in several research works [APBS24], [MCHJ18] and integrated into initial commercial applications, one exemplary solution being DeepFlow by CETAQUA [Ceta26], a Spanish center for water technology.

AI-based optimization models encompass performance, cost, and multi-objective optimization. One exemplary solution is the digital operating assistant for wastewater treatment plants developed by Nerou [Nero26], a company based in Cologne, which forecasts plant behavior by AI-based means and provides recommendations for the optimization of biological processes. Frequently investigated fields of application further include the prediction of influent volumes and loads [AnDB22], the optimization of energy-intensive processes such as aeration [LWHJ26], and predictive maintenance [XCYJ25]. These studies predominantly demonstrate that significant efficiency gains and environmental relief can be attained.

At the same time, a multitude of publications point out that the transfer into operational practice is associated with challenges [ICMB23], in particular, the availability and quality of suitable data [DIBH26], the integration into existing control systems, and the acceptance among operating personnel [WCLG23]. Questions regarding the transparency and traceability of AI decisions ("explainable AI") are likewise gaining importance within this safety-critical environment [WCLG23]. At the national level, the deployment of AI for wastewater treatment has hitherto been confined to a limited number of research projects and pilot installations, such as LiveSewer [RBFK25] and KIkKa [Plaz23]. The empirical evidence pertaining to the actual deployment of AI in German municipal wastewater treatment plants therefore remains deficient, in particular regarding the extent of routine use, the objectives pursued, and the barriers impeding wider dissemination.

3 Survey Methodology

The data collection was conducted by means of an anonymized online survey employing a non-representative sample. The questionnaire was developed in cooperation with the specialists of a wastewater treatment plant (current load size of approximately 171,000 design population equivalents (PE)) and subjected to repeated testing.

Wastewater treatment plants in Germany are classified into five size classes according to their population equivalents (PE), as defined in Annex 1 of the German Wastewater Ordinance [Fede04], based on a standard load of 60 g BOD5 per PE and day. This classification is commonly referenced by the German Association for Water, Wastewater and Waste (DWA). Class 1 plants serve up to 1,000 PE, Class 5 plants more than 100,000 PE, and Classes 2–4 cover intermediate capacities.

Of the approximately 8,900 public wastewater treatment facilities existing in Germany according to the Federal Statistical Office [BdDe24] the email addresses of employees from among the operational management were researched by means of web scraping and manual investigation. This resulted in 507 facilities, which were invited by email to the survey created with the software solution LamaPoll. 301 invitations were dispatched directly to facility employees.

The questionnaire combined closed, semi-open, and scaled questions. The general part encompassed the activity profiles of the respondents, structural company characteristics such as the respective DWA class, and wastewater-related key figures, the degree of digitalization, existing monitoring methods and control mechanisms, and past critical situations. The AI section ascertained the knowledge and deployment of AI-based solutions for technical processes of wastewater treatment, the potential benefits and the concerns towards such solutions, and the relevance of camera-based AI solutions.

4 Survey Results

The following, primarily descriptive analysis provides, by means of frequency analyses and cross-tabulations, an overview of the current status of AI deployment among the respondents. Although the results are not representative of all German public wastewater facilities, they provide indications of initial tendencies. Of the 507 facilities invited, 105 online questionnaires were completed in full. With 56 respondents from facilities serving 10,001 to 100,000 PE (DWA size class 4) and 39 from facilities serving more than 100,000 PE (DWA size class 5), the questions were predominantly answered by large wastewater facilities.

The respondents perform differing functions within their facilities, ranging from executive management and engineering to laboratory, operations, and IT. It may nonetheless be assumed that, they possess the knowledge relevant to the questionnaire. 83 individuals concur at least in part that the technical and organizational tasks pertaining to wastewater treatment exhibit a high degree of digitalization within their company.

4.1 Company-Wide Use of AI Applications

The survey results reveal a heterogeneous picture with respect to the company-wide use of AI-based applications. Most frequently, other fields of application were named (33 responses), which points to a broad yet imprecisely delineated use. 17 respondents employ AI in the control and/or monitoring of technical installations, and 16 for commercial or organizational tasks, whereas the measurement of wastewater quality has hitherto played a subordinate role (6 responses). It is furthermore noteworthy that 38 respondents possess no knowledge regarding the deployment of AI within their company, which points to a limited transparency and/or internal communication deficits.

4.2 AI Applications for Technical Processes of Wastewater Treatment

The respondents were asked to what extent AI-based solutions for the support and/or optimization of the technical process of wastewater treatment are known to them, or are even already in deployment within their company.

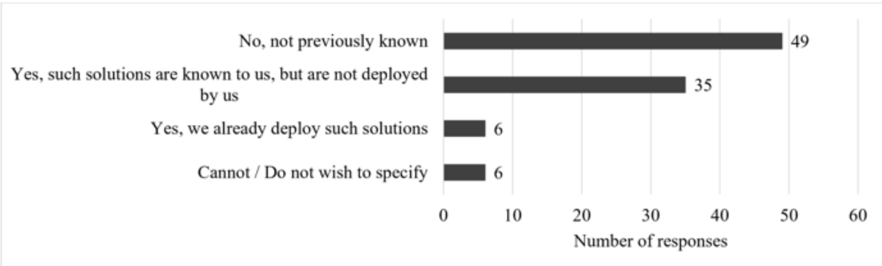

Fig. 1: Awareness and use of AI-based solutions for the support and/or optimization of the technical process of wastewater treatment.

Figure 1 illustrates the distribution of the responses. The largest group (49 respondents) reports that such applications have hitherto been unknown to them, which permits the inference of a distinct information deficit within the sector. 35 respondents are familiar with corresponding solutions yet have not implemented them, and merely 6 of the 96 facilities surveyed already actively employ such technologies. A further 6 respondents provided no information.

4.3 Potential Benefits of AI-Based Applications

The respondents perceive a multitude of potential benefits, whereby multiple responses were possible. Most frequently, the enhancement of operational safety is named (64 responses), followed by the improvement of the process stability (59 responses), the prevention or reduction of exceedances of limit values (46 responses), and improved documentation and reporting (42 responses). A reduction of manual work steps is anticipated by 34 respondents, whereas further aspects, such as the detection and attribution of problematic indirect discharges, the reduction of false alarms, and environmental aspects, are named less frequently. 10 respondents perceive no advantages.

4.4 AI-Based Monitoring of Critical Situations

Among the challenges faced by wastewater facilities is the timely identification of, and response to, critical situations such as blockages in the influent or sewer network, flooding, exceptional water temperatures, "invisible loads" caused by chemicals or viruses, and discernible loads such as foaming or discolorations. The survey therefore ascertained the process stages at which monitoring measures are carried out (multiple responses possible).

The monitoring predominantly takes place along several process stages: most frequently within the wastewater treatment process itself (76 responses), followed by influent points (52 responses) and direct discharges (48 responses), whereas indirect discharge points play a lesser role (29 responses). Overall, the findings indicate a predominantly established and multi-stage monitoring system.

Critical discharge situations at the influent points, such as fats, dyes, foam, suspended solids, or impermissible discharges are addressed with particular frequency in the literature [WCLG23]. According to the survey, such situations have hitherto occurred rather rarely: 77 of 104 respondents report extremely rare events (one to five cases per year), and a further 10 have never observed such situations, and more frequent occurrences remain the exception. Critical discharges thus do occur, yet do not constitute a regularly arising problem in most facilities. Important methods for the monitoring of the process flow are online measurements of wastewater parameters and automatic samplers. Camera-based AI systems such as DeepFlow [Ceta26] and LiveSewer [RBFK25], by virtue of their capacity for the real-time anomaly detection, may constitute an interesting complement. The respondents were therefore asked how important they deem such systems as complementary monitoring methods for certain critical situations.

The relevance is deemed particularly high for oil films, for which approximately 73% of respondents classify the detection as (very) important, followed by blockages (approximately 54%) and flooding (approximately 53%), whereas for foaming, suspended solids, turbidity, and discoloration a more balanced picture emerges. Immediately process-critical and potentially damage-relevant events thus stand at the focus of perception.

4.5 Barriers

Various research groups arrive, through empirical investigations, at the conclusion that the following barriers, above all, stand in the way of a comprehensive deployment of AI applications in wastewater facilities [ICMB23]: an insufficient availability of data, a lack of specialist competencies, integration effort, and economic uncertainties. In the study, potential barriers were addressed with a focus on camera-based AI detection solutions. It may nonetheless be assumed that the reported body of opinion may, at least in part, also be transferred to the general deployment of AI. The results demonstrate manifold concerns (multiple responses possible): most frequently, technical reliability and robustness are named (55 responses), followed by the integration into existing IT and infrastructure systems (49 responses), cost and economic-efficiency aspects (48 responses), requirements pertaining to critical infrastructures, in particular IT security (42 responses), and the implementation effort (38 responses). Further aspects are named less frequently, and merely 6 respondents harbor no reservations.

5 Data Analyses and Interview Results

In order to further address the research questions, the survey results are analyzed by means of bivariate analyses of possible relationships between company size and the assessment of potential benefits, fields of application, and barriers. Subsequently, selected insights from 10 expert interviews, each of half an hour in duration, contextualize and deepen the quantitative results.

5.1 Data Analysis

The following evaluations consider the size of the facility (DWA size class) and the prevailing level of awareness and use of AI-based solutions. The size-class analyses are confined to size classes 3 to 5, since the remaining classes were too sparsely represented for a meaningful group-wise interpretation.

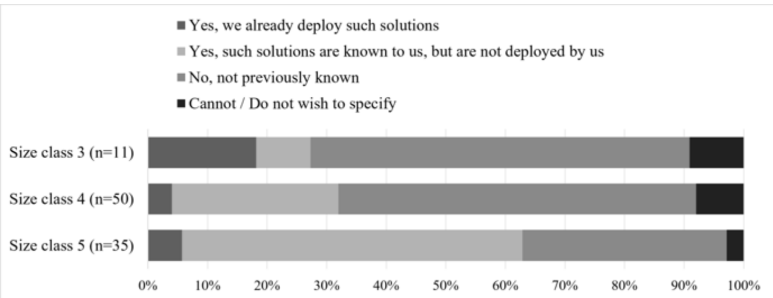

Fig. 2: Awareness and use of AI-based solutions for the technical processes of wastewater treatment, by DWA size class (percentage refers to the share of respondents within each size class; multiple responses possible).

As displayed in Figure 2, the response "not previously known" diminishes as facility size increases: from about 64% in size class 3 (n=11) and about 60% in size class 4 (n=50) to about 34% in size class 5 (n=35), whereas "known but not yet employed" rises markedly, from about 9% to about 57%, and represents the predominant category in size class 5.

Actual deployment ("already in use") remains limited throughout (approximately 4% to 18%). Awareness thus increases with facility size, whereas actual implementation remains low irrespective of size. With respect to the company-wide fields of AI application, a comparable picture emerges: in each class, a considerable proportion of respondents (about 33% to 42%) report having no knowledge of where AI is employed within their company, corroborating the limited internal transparency already observed in Section 4.1, while concrete operational applications remain modest irrespective of facility size. In all three classes, the enhancement of operational safety and the improvement of process stability constitute the most frequently named benefits (each about 50% to 60%), congruent with Section 4.3, and the prevention or reduction of exceedances of limit values likewise features prominently (about 41% to 45%). The core benefit expectations are thus shared across facility sizes, while more specific benefits display a size-dependent emphasis. In order to examine whether the assessment of potential benefits is contingent upon a facility's prior engagement with AI, the respondents were grouped according to their reported awareness and use of AI-based solutions (already in use, n=6; known but not employed, n=35; not previously known, n=49; no specification, n=6). The enhancement of operational safety constitutes the most prominently named benefit across the relevant groups (67% among those already deploying such solutions, 63% among those familiar but not yet deploying, and 76% among those not previously aware), closely followed by the improvement of process stability (67%, 69%, and 59%, respectively) and the prevention or reduction of exceedances of limit values (67%, 54%, and 47%, respectively). These core expectations are largely independent of prior exposure to AI. The group already employing such solutions (n=6) rates the majority of benefits particularly highly, although the very small size of this group necessitates a cautious interpretation. The perceived potential benefits are thus broadly shared, irrespective of the prevailing level of awareness or implementation.

5.2 Results of the Expert Interviews

The expert interviews reveal a differentiated picture: overall, AI is perceived as a relevant topic for the future. Yet practical deployment has hitherto been predominantly selective and concentrates on supporting applications. A central finding pertains to the underlying data: several experts regard insufficient data quality and a lack of standardization as substantial barriers, whilst one of the greatest potentials of AI is perceived in the automated processing and analysis of large volumes of operational data. Reference is likewise made to the need for uniform data models and for KRITIS- or NIS2-compliant interfaces (i.e. compliant with the German critical-infrastructure regulation and the corresponding EU directive). As relevant fields of application, process and operational optimization are named above all, including the optimization of energy consumption, the dosing of operating supplies, predictive maintenance, and anomaly detection. AI-based monitoring and control systems, for instance on the basis of digital twins or camera-based systems, are likewise assessed as fundamentally promising, and a municipal joint wastewater treatment plant already reports positive experiences with a digital twin for the intelligent control of stormwater overflow basins. At the same time, several experts emphasize that economic aspects and a comprehensible business case constitute decisive preconditions for adoption, and that municipal wastewater treatment plants, owing to their individual technical characteristics, are amenable to standardization only to a limited extent. In addition to technical challenges, organizational and human factors play an important role, including a lack of understanding of digital concepts, resistance to change, and concerns regarding possible job losses. Owing to the high responsibility for compliance with limit values under water law, the deficient traceability of many AI models (the "black-box problem") is regarded as a central barrier, and individual experts report erroneous decisions of learning-based systems in hitherto unknown operating states.

6 Discussion and Conclusion

The results indicate that the practical implementation of AI-based applications in German municipal wastewater treatment plants is still at an early stage, despite a widespread awareness of their potential. The findings identify structural, economic, and organizational barriers as the main reasons for this discrepancy, in particular technical reliability, integration into existing process control and IT systems, implementation and operating costs, as well as IT security and critical infrastructure requirements. At the same time, respondents expect AI solutions to provide significant benefits, particularly with regard to operational safety, process stability, efficiency, and improved documentation and reporting. The multi-stage monitoring systems already well established in wastewater treatment plants offer favorable conditions for the integration of AI-supported technologies. The bivariate analyses further reveal that awareness of AI increases with plant size, whereas actual implementation remains consistently low across all size classes. The expected benefits, in contrast, are perceived similarly regardless of plant size, prior AI experience, or the occurrence of critical operational events. Based on these findings, future efforts should focus on practical demonstration projects, transparent cost-benefit analyses, and the development of robust, user-friendly AI solutions that can be integrated into existing infrastructures with minimal effort. In addition, targeted training and knowledge transfer are essential to improve staff acceptance and foster confidence in AI technologies. The study is subject to several limitations, including the relatively small and non-representative sample, potential biases resulting from self-reported responses, and varying levels of respondents' knowledge about AI. Furthermore, some subgroup analyses are based on only a limited number of observations and should be interpreted as descriptive and exploratory rather than statistically generalizable. Future research should validate specific use cases under real operating conditions and provide robust assessments of their economic viability. In the long term, AI is expected to become a key enabler of the digital transformation of the sector, with the large volumes of archived operational and process data already available at most treatment plants representing a valuable foundation for training domain-specific AI models and accelerating the development of practical AI applications.

Bibliography

[AnDB22] ANDREIDES, MARKÉTA ; DOLEJŠ, PETR ; BARTÁČEK, JAN: The prediction of WWTP influent characteristics: Good practices and challenges. In: *Journal of Water Process Engineering* Vol. 49 (2022), p. 103009

[APBS24] ANTONINI, GIORGIO ; PEARCE, JOSHUA M. ; BERRUTI, FRANCO ; SANTORO, DOMENICO: A novel camera-based sensor for real-time wastewater quality monitoring. In: *Water Practice & Technology* Vol. 19 (2024), No. 9, pp. 3778–3793

[BdDe24] BDEW ; DESTATIS: *Abwasserdaten Deutschland, Strukturdaten der Abwasserentsorgung*. Berlin : BDEW Bundesverband der Energie- u. Wasserwirtschaft e.V., 2024. URL https://www.bdew.de/media/documents/BDEW_Broschuere_Abwasserdaten_2025_6Aufl_17072025_1.pdf. - accessed on 2026-04-20

[BJBB22] BÜTTNER, OLAF ; JAWITZ, JAMES W. ; BIRK, SEBASTIAN ; BORCHARDT, DIETRICH: Why wastewater treatment fails to protect stream ecosystems in Europe. In: *Water Research* Vol. 217 (2022), p. 118382

[Ceta26] CETAQUA: Detection of hydrocarbons in wastewater and rivers using cameras installed at critical points. Barcelona : CETAQUA Water Technology Centre, 2026. URL https://www.cetaqua.com/en/proyectos/deepflow-eng/ - accessed on 2026-04-25

[DIBH26] DE LA HOZ, JEIMY MARTINEZ ; ISLAM, M. D. SHAFIKUL ; BAPPY, MAHATHIR MOHAMMAD ; HAYES, MICHAEL P.: AI-enabled modeling for smart rural wastewater treatment systems: current practices and remaining gaps. In: *Applied Water Science* Bd. 16 (2026), No. 2, p. 40

[Euro23] EUROPEAN COMMISSION, DIRECTORATE-GENERAL FOR ENVIRONMENT: *Are waste-water treatment plants failing to protect the ecological health of European streams?, 2023-02-15*. URL https://environment.ec.europa.eu/news/are-waste-water-treatment-plants-failing-protect-ecological-health-european-streams-2023-02-15_en. - accessed on 26-06-26 — Science for Environment Policy

[Fede04] FEDERAL MINISTRY OF JUSTICE (GERMANY): *Anhang 1 AbwV – Häusliches und kommunales Abwasser [Annex 1 AbwV – Domestic and Municipal Wastewater]*. URL https://www.gesetze-im-internet.de/abwv/BJNR056610997.html. - accessed on 2026-06-26. — Gesetze im Internet

[HRGM25] HOCAOGLU, SELDA MURAT ; ROGHANI, BARDIA ; GULCAN, HANDE ; MAGNA, DÉBORA JARETA ; AYDÖNER, CIHANGIR ; BARROS, VIRGINIA GRACE ; KOYUNLUOGLU AYNUR, SEBNEM ; RATNAWEERA, HARSHA ; ET AL.: Artificial intelligence in wastewater treatment plants: a review of current trends and adaptation strategies in the face of climate-driven rainfall fluctuations. In: *Journal of Water and Climate Change* Vol. 16 (2025), No. 9, pp. 2742–2759

[ICMB23] IMEN, SANAZ ; CROLL, HENRY C. ; MCLELLAN, NICOLE L. ; BARTLETT, MARK ; LEHMAN, GENO ; JACANGELO, JOSEPH G.: Application of machine learning at wastewater treatment facilities: a review of the science, challenges and barriers by level of implementation. In: *Environmental Technology Reviews* Vol. 12 (2023), No. 1, pp. 493–516

[LWHJ26] LI, QUNLI ; WANG, KAI ; HORACE, GENEVIEVE FAITH ; JIANG, PEITING ; XIONG, YANYAN

; HUANG, FUJIAN ; WANG, YONG ; XU, WENLAI ; ET AL.: Research on an intelligent precise aeration control system for wastewater treatment based on LSTM models. In: *Journal of Water Process Engineering* Vol. 85 (2026), p. 109746

[MCHJ18] MULLINS, DARRAGH ; COBURN, DEREK ; HANNON, LOUISE ; JONES, EDWARD ; CLIFFORD, EOGHAN ; GLAVIN, MARTIN: A novel image processing-based system for turbidity measurement in domestic and industrial wastewater. In: *Water Science and Technology* Vol. 77 (2018), No. 5, pp. 1469–1482

[Nero26] NEROU: nerou elements. Die moderne Steuerung von Kläranlagen. Köln : nerou GmbH, 2026. URL https://www.nerou.de/landing/produkt.html. - accessed on 2026-04-25

[Plaz23] PLAZZO, MICHAELA: BMUV fördert KI-Leuchtturmprojekt zu klimaneutralen Kläranlagen. In: *EUWID Wasser und Abwasser* Vol. 2023. Gernsbach (2023), No. 28. URL https://www.euwid-wasser.de/news/wirtschaft/bmuv-foerdert-ki-leuchtturmprojekt-zu-klimaneutralen-klaeranlagen-290623/. - accessed on 26-06-26

[RBFK25] ROHRBACH, MICHAELA ; BÖDECKER, JÖRG ; FUHRMANN, TIM ; KLÖCKER, PAULA ; LAU, PHILIP ; ZIMMERMANN, MARTIN: Kanalmonitoring mit KI - vom Potenzial zur Umsetzung. In: *gwf-Wasser/Abwasser International* Vol. 11. Essen, Vilkan Verlag (2025), No. 166, pp. 65–73

[ShAH20] SHARAFATI, AHMAD ; ASADOLLAH, SEYED BABAK HAJI SEYED ; HOSSEINZADEH, MAJID: The potential of new ensemble machine learning models for effluent quality parameters prediction and related uncertainty. In: *Process Safety and Environmental Protection* Vol. 140 (2020), pp. 68–78

[WCLG23] WANG, YI ; CHENG, YUHAN ; LIU, HE ; GUO, QING ; DAI, CHUANJUN ; ZHAO, MIN ; LIU, DEZHAO: A Review on Applications of Artificial Intelligence in Wastewater Treatment. In: *Sustainability* Vol. 15 (2023), No. 18, p. 13557

[XCYJ25] XIONG, ZHAOYANG ; CHENG, LEIMING ; YUNUS, AHMED IBRAHIM ; JIAN, DEWU ; ZHANG, XINPING ; IGOU, THOMAS ; TONG, SHA ; CHEN, YONGSHENG: Machine Learning-Assisted Expert Control in Wastewater Aeration Treatment Process Application. In: *ACS ES&T Engineering* Vol. 5 (2025), No. 11, pp. 3140–3152